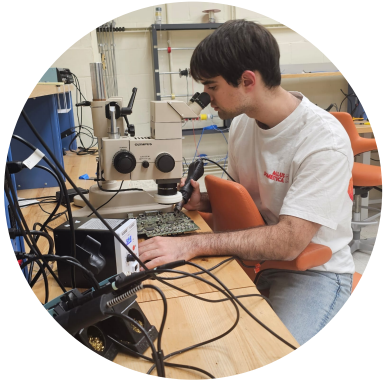

Andres Torrubia

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The future is here. There are still orders of magnitude to unlock by engineering the full stack, from electromagnetism and silicon to systems and APIs.

EDUCATION

Duke University
Triple Major in

- BSE Electrical and Computer Engineering
- BS Mathematics
- BS Computer Science

2023 - 2027
GPA 3.89

SKILLS

Programming Languages
Vibecoding, C/C++, Rust, CUDA, Python, Verilog, MIPS/RISCV assembly.

Languages
English (Native), Spanish (Native), Mandarin (C1).

Software
Proficient
PyTorch, NumPy, OpenMP, Git, Linux/Unix, Arduino, KiCad.
Working Knowledge
LTspice, Docker, Xilinx Vivado, GNU/LLVM toolchain.

Practical
Agent orchestration, PCB design, prototyping, soldering, data-sheet and ISA reading.

EXPERIENCE

Atomic Semi
Guardian of the Psi Delta, Intern Jun–Aug 2025 and Summer 2026

- Rebuilt the company's ellipsometry mathematics and software stack from the ground up, making the system roughly 3000x faster.
- Returning in Summer 2026 to work on scatterometry.
- Was offered a full-time role.

Metamaterial Research
Research (with Prof. David R. Smith) Current

- Currently developing GPU-accelerated Discrete Dipole Approximation algorithms to simulate and control electromagnetic responses for 3d metamaterials.
- Invited to contribute a paper to *Nature Communications*.
  The work is going well and David has invited us to do a PhD.

Kaggle Silver Medal, GoogleAI4Code
Deep Learning Competition Summer 2022

- Won a silver medal by training transformers and developing a differentiable Kendall Tau style loss for the ranking objective.

Recommendation Systems and Image Generation
Intern, Magnific Summer 2022 and Summer 2023

- Developed a transformer-based recommendation system trained autoregressively, used Faiss for similarity search and multi-GPU for fast training and inference.
- Engineered an image dataset generation pipeline from public domain media using FFmpeg and captioning models. Validated the dataset's efficacy by fine-tuning diffusion models with LoRA.

Other Projects
Selected software, hardware, and research builds 2024–Present

- **FPGA Raycaster Engine**, built a custom pipelined CPU on FPGA and then built a bare-metal assembly raycaster on top of it, tuning the memory architecture for real-time performance.
- **LightLemur**, leading three people building a lightweight tensor differentiation library with OpenMP and SIMD, achieving near-PyTorch CPU performance.
- **HummingBird**, built a CUDA attention kernel for short sequences that ran nearly 3x faster than FlashAttention on consumer GPUs.
- **Penguin**, vibe-coded my own transcription pipeline that runs faster than WhisperFlow, stays fully local and private, and is used by my friends and family.
- **Hardware**, built an analog AM radio from scratch using datasheets and fundamentals and designed a 110 W power converter PCB with active PFC and an LLC resonant DC-DC stage.
- **Duke Hacker Fab**, started the Duke Hacker Fab and am currently building a lithography stepping machine for practical semiconductor tooling.